SCOPE OF WORK

Where Has the Money Gone?

U.S. Labor Market & Industry Distribution Analysis

Program	American Economic Data Repository (AEDR)
Project Type	Independent economic data analytics / portfolio project
Analytical Baseline	National 2-digit NAICS sectors
Primary Period	1990-2025
Workflow Standard	Ask -> Prepare -> Process -> Analyze -> Share -> Act

Purpose

Define the work, controls, deliverables, acceptance criteria, and analytical workflow for a reproducible labor-market analysis that demonstrates professional data-analytics practice from problem framing through project closeout.

1. Project Overview

Where Has the Money Gone? is the first major analytical project built from the American Economic Data Repository (AEDR). AEDR is the underlying public economic-data infrastructure used to preserve, validate, organize, and document source data for repeatable research rather than one-off analysis.

This project examines how employment, wages, and economic gains have been distributed across the U.S. labor market over time. The baseline is intentionally broad: national 2-digit NAICS sectors are used to establish the structural labor-market picture before any finer industry, ownership, geographic, occupational, diagnostic, or predictive drill-down is considered.

The work is one component of a larger American Dream Affordability research program. The labor-market project establishes the economic-opportunity side of affordability; a subsequent housing-affordability project will examine housing costs relative to income and economic conditions; a later integrated project will evaluate where employment opportunity and housing affordability combine most favorably and will inform development of an American Dream Affordability Index.

Primary research question
Where have the gains in wages, employment, and economic opportunity gone within the U.S. labor market?

2. Business / Research Need

Headline labor-market measures can show whether payrolls or earnings are rising in aggregate, but they do not by themselves explain how those gains are distributed across major parts of the economy. This project translates that broad question into measurable industry-level comparisons using long-run employment, wage, and inflation-adjusted compensation data.

The analytical objective is not to prove a predetermined conclusion. Success is defined as producing a valid, reproducible, well-documented answer - including a mixed or unexpected answer - and clearly distinguishing measured evidence from interpretation.

3. Stakeholders and Intended Audience

Audience / Role	Need	Project Response
Project owner / analyst	A disciplined framework for executing and documenting the analysis.	Reproducible notebook, validated analytical data, scope controls, findings, and limitations.
Technical reviewers / analysts	Ability to inspect assumptions, grain, transformations, metrics, and validation.	Transparent methods, source context, analytical checks, and documented limitations.
Recruiters / hiring managers	Evidence of real-world analytical workflow and judgment, not only finished charts.	Scope of work, analytical framework, notebook, documentation, and public repository structure.
Public research users	Clear interpretation of long-run labor-market patterns.	Accessible findings with source provenance, definitions, and cautions.

4. Scope Definition

4.1 In Scope - Baseline Project

- National U.S. labor-market analysis using broad 2-digit NAICS sectors as the baseline industry classification.
- Long-run QCEW coverage from 1990 through 2025, subject to documented historical comparability limitations.
- Employment levels and sector employment shares.
- Total annual wages and sector wage shares.
- Average annual pay, including nominal and CPI-adjusted real-pay comparisons.
- Long-run changes in employment share, wage share, and real average annual pay.
- Annual employment redistribution and wage redistribution across broad sectors.

- Wage-versus-employment redistribution gaps and sector-year divergence analysis.
- Data validation, aggregation checks, missing/suppressed-cell handling, historical-comparability review, and reproducible documentation.
- Clear separation of descriptive evidence, association, interpretation, and unsupported causal claims.

4.2 Optional Drill-Downs

Optional extensions are initiated only when a baseline finding creates a defined analytical reason to add detail. Availability of more granular data alone is not sufficient justification.

- Finer NAICS analysis: subsectors, industry groups, industries, or national industries.
- Ownership analysis: private, federal, state, and local government where source structure supports comparison.
- Geographic analysis: state, CBSA, or county patterns and industry-by-geography comparisons.
- Occupational analysis using OEWS when the question requires distinguishing industry-level compensation from occupational outcomes.
- Diagnostic or explanatory analysis using relevant macroeconomic or labor-market variables.
- Predictive analysis only when a defensible target, sufficient history, and time-aware validation plan are defined.

4.3 Out of Scope for the Baseline

- Claims that the descriptive analysis establishes causation.
- A requirement to analyze every lower-level NAICS, ownership, geography, or occupation combination.
- A predetermined machine-learning model or requirement to use machine learning.
- Construction of the final American Dream Affordability Index within this project.
- Housing-affordability analysis beyond contextual references to the larger research program.
- Prescriptive policy recommendations unless a separate decision context and supporting evidence are defined.
- Adding alternative wage series solely for redundancy when they do not answer a new analytical question.

5. Objectives and Success Criteria

Objective	Measure of Completion	Acceptance Standard
Define the analytical grain	National 2-digit NAICS sector-year baseline documented.	Every core metric can be traced to a consistent observation level.
Validate the analytical data	Coverage, uniqueness, aggregation, null/suppression, and historical checks completed.	No known double counting; exceptions and limitations documented.
Measure compensation in context	Nominal and real average annual pay derived and compared.	Inflation adjustment is explicit and reproducible.
Measure distributional change	Employment share, wage share, redistribution, and divergence metrics produced.	Metrics use appropriate denominators and are independently sanity-checked.
Communicate defensibly	Findings, limitations, and follow-up questions documented.	Claims do not exceed what the evidence supports.
Preserve reproducibility	Notebook, analytical inputs, documentation, and versioned outputs retained.	A reviewer can follow the workflow and understand how results were produced.

6. Data Requirements and Source Controls

Source	Role	Coverage	Primary Measures	Control
BLS QCEW / AEDR analytical Parquet	Primary labor-market source	1990-2025	Employment, establishments, total wages, average weekly wage, average annual pay, ownership, geography, industry	Validate grain, sector mapping, aggregation behavior, missing/suppressed cells, and historical comparability.
CPI / AEDR macro data	Inflation adjustment	Aligned to analytical period	Price level / inflation index	Use a documented base/reference approach and preserve nominal values alongside real measures.

NAICS reference hierarchy	Industry classification	Applicable classification structure	Sector codes and titles	Validate combined 2-digit sector identifiers and classification meaning.
BLS documentation	Method and source interpretation	As applicable	Definitions, aggregation, reconstruction, publication conventions	Use to interpret source structure and document limitations.

6.1 Data Handling Principles

- Preserve source-derived data separately from transformed analytical outputs whenever practical.
- Maintain provenance from published source through SQLite / analytical views and Parquet outputs to notebook-derived metrics.
- Do not convert absent, unavailable, suppressed, not-applicable, and true-zero observations into one undifferentiated condition.
- Join or aggregate only after confirming compatible keys, grain, and cardinality.
- Document source changes and historical reconstruction instead of manufacturing a falsely uniform panel.

7. Work Plan - Ask -> Prepare -> Process -> Analyze -> Share -> Act

The project workflow follows the six-phase analytical process used in the Google Data Analytics framework and adapts it to the AEDR research environment. Each phase has a defined purpose, work product, and completion gate.

ASK - Frame the problem before analysis

- Define the research question and intended audience.
- Set the national 2-digit NAICS baseline.
- Define in-scope and out-of-scope work.
- Define success as a valid, reproducible answer rather than a preferred conclusion.

Phase output: Approved scope, research questions, success criteria, analytical boundaries.

PREPARE - Acquire and understand the right data

- Inventory source datasets and documentation.
- Confirm coverage, frequency, observation grain, keys, and provenance.
- Assess reliability, traceability, completeness, and historical comparability.
- Establish file organization, naming, and version control.

Phase output: Source inventory, data requirements, provenance notes, initial data profile.

PROCESS - Clean, transform, and validate

- Profile rows, columns, types, dates, codes, and ranges.
- Check missing/suppressed values and duplicates at the intended key.
- Validate NAICS mappings and aggregation behavior.
- Inspect invalid combinations, extreme values, and structural breaks.
- Validate processed outputs against source controls and known identities.
- Record exceptions and methodological decisions chronologically.

Phase output: Validated analytical dataset, transformation logic, quality checks, audit record.

ANALYZE - Turn validated data into evidence

- Establish national sector employment and wage baselines.
- Calculate nominal and real average annual pay growth.
- Calculate employment and wage shares and long-run changes.
- Measure annual redistribution and wage-versus-employment divergence.
- Investigate notable periods or sectors without converting association into causation.
- Recalculate or sanity-check important metrics independently where practical.

Phase output: Reproducible notebook, derived metrics, tables/figures, findings, limitations.

SHARE - Communicate for the intended audience

- Lead with the question, evidence, finding, and limitation.
- Use clear tables or visuals without exaggerating magnitude.
- Keep technical methods available in the notebook and documentation.
- Publish portfolio-facing documentation that explains data, methods, outputs, and reproducibility.

Phase output: GitHub repository, README, scope of work, analytical framework, notebook, selected visuals / tables.

ACT - Close the loop

- State what the evidence supports and does not support.
- Separate follow-up questions from the completed baseline scope.
- Archive final data products, code, documentation, and version history.
- Identify justified next analyses: finer industry, ownership, geography, occupation, or the next affordability project.

Phase output: Documented conclusion, archived deliverables, follow-up backlog, project handoff / next-step record.

8. Deliverables

Deliverable	Purpose	Format	Status / Standard
Scope of Work	Defines boundaries, controls, workflow, deliverables, and acceptance criteria.	DOCX / repository document	Portfolio-ready; aligned to project execution.
Analytical & Research Framework	Defines AEDR context, research logic, baseline grain, analytical stages, and optional drill-downs.	DOCX / PDF	Maintained as methodological context.
QCEW Analytical Notebook	Documents preparation, validation, derived metrics, analysis, and interpretation.	Jupyter Notebook	Reproducible and reviewable.
AEDR Analytical Data	Provides validated labor-market and macro inputs used by the notebook.	Parquet / SQLite-derived analytical outputs	Source provenance and schema documented.
Audit / Engineering Record	Preserves validation findings, anomalies, decisions, and corrections.	Chronological documentation	Exceptions recorded without rewriting history.
README / Project Brief	Explains the problem, data, methods, major outputs, limitations, and reproduction path.	Markdown	Designed for rapid review by technical and nontechnical audiences.
Selected Visuals / Tables	Communicates decision-relevant findings clearly.	Notebook / Tableau / image outputs	Accurate labels, units, dates, denominators, and non-distorting scales.

9. Milestones and Completion Gates

Gate	Minimum Completion Standard	Baseline Project Status
ASK	Problem, audience, questions, success criteria, and scope documented.	Complete
PREPARE	Sources inventoried; grain, provenance, coverage, and storage structure established.	Complete
PROCESS	Quality checks completed; transformations reproducible; exceptions and source limitations documented.	Complete
ANALYZE	Metrics match questions; benchmarks and assumptions explicit; results checked and interpreted conservatively.	Complete
SHARE	Portfolio documentation, notebook, and public-facing repository materials prepared.	Complete

ACT	Final deliverables archived; conclusions documented; optional follow-ups separated from baseline scope.	Complete

10. Quality Assurance and Acceptance Criteria

- Analytical grain is explicitly defined before calculation, aggregation, or comparison.
- No aggregate ownership category is double-counted with its component categories.
- 2-digit NAICS sector mappings, including combined sector identifiers, are validated.
- Duplicate checks are performed at the intended analytical key.
- Missing, suppressed, unavailable, structurally absent, and true-zero observations are not conflated.
- Inflation-adjusted measures retain a traceable relationship to nominal source measures and the selected CPI series.
- Historical source limitations - including reconstructed pre-2001 NAICS observations - are disclosed.
- Derived shares use denominators that match the research question.
- Important metrics receive reasonableness checks or independent recalculation where practical.
- Findings distinguish descriptive evidence and association from causal explanation.
- All transformations, exclusions, assumptions, exceptions, and material scope changes are documented.

Acceptance rule

A result is not accepted because it supports an expected narrative. It is accepted when the source, grain, transformation, calculation, and interpretation can be defended and reproduced.

11. Assumptions, Constraints, and Risks

Item	Project Implication	Control / Mitigation
Historical classification changes	Long-run comparisons may contain additional comparability uncertainty.	Preserve and disclose BLS reconstruction / classification notes; avoid overstating precision.
Published aggregate structure	Mixing totals with component records can double count employment or wages.	Validate aggregation levels and ownership structure before analysis.
Suppression / absent source cells	Blank or absent combinations may not represent zero economic activity.	Keep conditions distinct and avoid automatic zero-filling.
Average annual pay is an aggregate measure	It does not represent the median worker or every individual's earnings experience.	Use precise terminology and avoid interpreting sector averages as individual outcomes.
Descriptive design	Observed divergence does not identify a causal mechanism.	Use causal language only if a separate defensible design is developed.
Scope expansion	Large AEDR coverage can encourage analysis that is not necessary to answer the baseline question.	Require a defined finding or research question before initiating drill-downs.

12. Change Control and Documentation

Scope changes, methodological corrections, unresolved anomalies, and engineering decisions are recorded as new chronological entries. Earlier records are not retroactively rewritten to make the project history appear cleaner. When a later finding changes an earlier interpretation, the later entry references and amends or supersedes the earlier record.

A proposed analytical extension is treated as a scope change when it introduces a new population, geography, data source, target variable, causal claim, modeling objective, or required deliverable. Optional drill-downs are documented separately so they do not silently become requirements of the completed baseline.

13. Roles and Responsibilities

Role	Responsibility
Project owner / analyst	Define scope; acquire and validate data; maintain AEDR structure; execute analysis; document decisions; interpret results; prepare public deliverables.
Source agencies / publishers	Provide authoritative source data, definitions, classifications, and methodological documentation.
Technical reviewer / future collaborator	Review reproducibility, assumptions, calculations, limitations, and proposed extensions where applicable.
Portfolio audience	Evaluate the project as evidence of analytical workflow, technical execution, documentation, and communication.

14. Project Closeout and Handoff

The baseline project is considered closed when the validated analytical workflow, notebook, findings, limitations, and portfolio documentation are archived in the public project repository and the remaining drill-down questions are clearly separated as future work.

- Archive final notebook and reproducible analytical inputs.
- Publish the scope of work and analytical framework alongside the project.
- Publish a concise README explaining the research question, data, methods, findings, limitations, and reproduction path.
- Retain audit / validation documentation as evidence of data-quality controls.
- Identify follow-up analyses without expanding the completed baseline retroactively.
- Carry relevant labor-market findings forward into the later housing-affordability and integrated affordability projects.

15. Professional Workflow Summary

Phase	Professional Question	Evidence Produced
Ask	What problem are we responsible for answering?	Scope, research question, success criteria, boundaries.
Prepare	Do we have the right data, at the right grain, with traceable provenance?	Source inventory, coverage, keys, data profile.
Process	Can the analytical dataset be trusted?	Validation checks, transformations, audit record, exceptions.
Analyze	What does the validated evidence show?	Metrics, comparisons, trends, relationships, limitations.
Share	Can the intended audience understand and inspect the work?	README, notebook, framework, visuals, technical detail.
Act	What does the evidence support, and what happens next?	Documented conclusion, archived deliverables, separated follow-up work.

Guiding principle: The model is not the project.

The project is the disciplined use of economic data to frame a question, establish trustworthy evidence, apply methods appropriate to the task, communicate limitations, and preserve a reproducible path from source to conclusion.

Method Basis

This Scope of Work applies the analytical framework developed for the American Economic Data Repository (AEDR) within the **Ask → Prepare → Process → Analyze → Share → Act** workflow taught through the **Google Data Analytics Professional Certificate**. The six-phase process provides the organizational foundation for defining the problem, preparing and validating data, conducting analysis, communicating results, and translating findings into appropriate next steps.